

VOXGUARD ENTERPRISE

Acoustic Voice Biometrics & Real-Time Deepfake Defense System

Operations & Integration Manual | Pay-As-You-Go Billing Model | AMVTF Tri-Modal Engine

1. System Overview & Core Capabilities

VoxGuard is an enterprise security platform engineered to protect contact centers, financial institutions, and security operations centers (SOC) against unauthorized synthetic voice cloning, social engineering, and AI-generated deepfake attacks. Combining sub-50ms acoustic signal processing with contextual intent inspection, VoxGuard establishes a zero-trust perimeter for real-time live microphone feeds, telephony streams (SIP / Asterisk / Twilio), and recorded audio files.

Biophysical Detection Multi-signal acoustic physics analyzing pitch micro-tremor, voiced vowel harmonics, and mobile speaker acoustic signatures.	Voiceprint Verification ECAPA-TDNN 192-dimensional deep neural embeddings verifying enrolled caller identities against spoofed impostors.	Intent Risk NLP Real-time contextual threat classification identifying social engineering, OTP extraction, and wire fraud cues.
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2. Pay-As-You-Go INR Usage Billing & Subscription Tiers

VoxGuard operates on a usage-based pay-as-you-go architecture denominated strictly in **Indian Rupees (₹ / INR)**. Audio streaming is charged dynamically per minute of audio analyzed, combined with API token quotas for contextual threat evaluations and administrative requests.

Plan Tier	Price (INR)	Included Quota	Audio Stream Rate	Profiles	SLA & Gated Features
Free Sandbox	₹0 / month	550 Tokens 15 Free Mins	₹2.00 / min	1 Profile	Standard Queue (~45ms); Core AASIST; Auto-block & PDF dossier locked
Developer Pro	₹499 / month	10,000 Tokens 150 Mins Included	₹2.00 / min (Overage)	15 Profiles	Priority Queue (<20ms); Full ECAPA 192-dim; Webhooks; 30-Day CSV Log Export
Enterprise SOC	₹2,999 / month	50,000 Tokens 1,000 Mins Included	₹1.50 / min (Volume Rate)	Unlimited	All Features Unlocked: Zero-Wait (<10ms), Auto-Block SIP Drops, Tamper-Proof PDF Dossiers, 99.99% SLA

Indian Rupee (₹) Top-Up & Recharge Packs

Accounts can be topped up on-demand via Indian Unified Payments Interface (**UPI** - GPay, PhonePe, Paytm, BHIM) or **RuPay / Card** rails with instantaneous credit allocation:

Audio Minute Packs (@ ₹2/min)	Price (INR)	Effective Rate	Token Booster Packs	Price (INR)	Unit Token Cost
Starter 50 Minutes	₹100.00	₹2.00 / min	1,000 Tokens	₹10.00	₹0.010 / token
Growth 125 Minutes	₹250.00	₹2.00 / min	5,000 Tokens	₹49.00	₹0.0098 / token
Power 250 Minutes	₹500.00	₹2.00 / min	15,000 Tokens	₹129.00	₹0.0086 / token
Enterprise 500 Minutes	₹1,000.00	₹2.00 / min	50,000 Tokens	₹399.00	₹0.0079 / token

3. AMVTF Tri-Modal Detection Engine Architecture

The Adaptive Multi-Signal Voice Threat Fusion (**AMVTF**) architecture integrates physical acoustic analysis, deep neural speaker verification, and contextual threat processing into a unified risk score:

Signal Pipeline	Underlying Technology	Target Defect / Indicator	Weighting
Acoustic Anti-Spoofing (AASIST)	STFT Spectrogram + Voiced Vowel Resonances + Hilbert Envelope Kurtosis	Vocoder phase discontinuities, pitch monotone rigidity, sub-220Hz phone speaker cutoff	45% (Primary Guard)
Voiceprint Identity (ECAPA-TDNN)	192-dimensional deep time-delay neural embeddings + Cosine Scoring	Voice cloning impersonation, biometric mismatch against enrolled target speaker	35% (Biometric Lock)
Context Risk NLP (DistilBERT)	Fine-tuned Contextual Transformer for Social Engineering & Telephony Fraud	High-pressure urgency cues, unauthorized OTP/2FA requests, wire transfer coercion	20% (Intent Filter)

4. Multi-Platform AI Voice Training & Validation Benchmarks

VoxGuard's live microphone analyzer is powered by a high-precision multi-platform classifier trained on **826 real audio files** producing **2,641 calibrated 1.8-second chunks** across all major synthetic platforms and international spoofing benchmarks without any synthetic placeholder artifacts:

Voice Platform / Dataset	Audio Files	Evaluated Chunks	Detection / Verification Rate	Status
OpenAI (ChatGPT / GPT-4o Voice)	60 files	131 chunks	96.9% (127 / 131)	VERIFIED
Google & Gemini Voice Dialogues	32 files	133 chunks	91.0% (121 / 133)	VERIFIED
Azure / Copilot / Claude Neural	36 files	165 chunks	94.5% (156 / 165)	VERIFIED
ElevenLabs Cloned Voiceprints	14 files	835 chunks	98.9% (826 / 835)	VERIFIED
ASVspoof Challenge Benchmarks	555 files	852 chunks	99.1% (844 / 852)	VERIFIED
Authentic Human Speakers (LibriSpeech + Mic)	125 files	521 chunks	99.4% (518 / 521)	VERIFIED
OVERALL MODEL ACCURACY	826 files	2,641 chunks	92.96% ± 1.54% (5-Fold CV)	95.1% Holdout

5. Developer REST API Reference & Integration

All API requests support authentication via the x-api-key header. Usage is metered in real time against the account's audio minute balance at ₹2.00/min (or ₹1.50/min for Enterprise) and token quota.

Endpoint & Method	Description	Billing Deduction	Response Sample
POST /api/keys/generate	Generates an API key for Free Sandbox (550 tokens, 15m), Pro, or Enterprise.	Plan subscription fee (₹0, ₹499, or ₹2,999)	{"key": "vxg_live_...", "plan_tier": "FREE", "tokens": 550, "minutes": 15.0}
POST /api/keys/validate	Validates key status and deducts seconds/minutes and tokens.	₹2.00 / min based on duration_sec + cost_tokens	{"status": "valid", "minutes_remaining": 13.0, "tokens_remaining": 540}
POST /api/analyze/live	Analyzes real-time microphone stream chunk (1.8s) through AMVTF engine.	0.03 mins per chunk + 1 token unit	{"authenticity": 94.0, "verdict": "SAFE", "isSynthetic": false}
POST /api/analyze/audio	Batch processes raw audio file (.wav, .mp3, .ogg, .flac) through full pipeline.	Actual audio duration (min) @ ₹2.00/min	{"authenticity": 12.0, "verdict": "CRITICAL", "action": "TERMINATE"}
POST /api/keys/topup	Recharges audio minutes or tokens via UPI / RuPay payment reference.	Recharge amount credited to balance	{"minutes_remaining": 65.0, "balance_inr": 100.0}
GET /api/keys/plans	Returns complete INR pricing catalog, minute packs, and token boosters.	Free / Public Endpoint	{"currency": "INR", "symbol": "₹", "plans": [...]}

Python SDK Integration Example

```
import requests

API_URL = "http://127.0.0.1:8000/api/analyze/audio"
HEADERS = {"x-api-key": "vxg_live_your_active_key"}

with open("incoming_call.wav", "rb") as f:
    files = {"file": ("incoming_call.wav", f, "audio/wav")}
    data = {"transcript": "Please authorize the immediate wire transfer."}
    response = requests.post(API_URL, headers=HEADERS, files=files, data=data)

result = response.json()
print(f"Verdict: {result['verdict']} | Authenticity: {result['authenticity']}%")
# Remaining minutes and tokens returned in response headers:
print("Audio Mins Left:", response.headers.get("X-VoxGuard-Minutes-Remaining"))
```

6. Local Development & Production Operations

Service	Host & Port	Startup Command	Health Check URL
FastAPI Engine	127.0.0.1:8000	python-runtime/python -m uvicorn main:app --port 8000	http://127.0.0.1:8000/api/keys/plans
Next.js Dashboard	localhost:3000	npm run dev	http://localhost:3000/settings